Informe - Exercici 3: Compressio GIF

Assignatura: Tractament d'imatges
Data de generacio: 2026-02-18

Aquest informe resumeix la conversio de les imatges PNG a GIF, l'analisi de paleta de colors, el calcul de bytes per pixel i l'efecte visual d'anul-lar la component vermella (R=0) en les imatges 3, 7 i 8.

Resum numeric (apartats c i e)

Fitxer	Colors GIF	B/px GIF	Colors PNG	B/px PNG
image1.gif	256	1.000	55117	3.000
image2.gif	256	1.000	63558	3.000
image3.gif	256	1.000	45558	3.000
image4.gif	256	1.000	25567	3.000
image5.gif	256	1.000	24470	3.000
image6.gif	256	1.000	72079	3.000
image7.gif	256	1.000	258	3.000
image8.gif	244	1.000	2377	3.000

Comparacio (apartat f)

El GIF fa servir una paleta indexada de fins a 256 colors i acostuma a 1 byte/pixel. El PNG RGB manté molta mes variacio cromatica i fa servir 3 bytes/pixel sense compressio. Visualment, en GIF poden apareixer perdues en gradients i textures fines per la quantitzacio de color.

Detall de image1.png

GIF: image1.gif | Colors: 256 | B/px: 1.000
PNG: image1.png | Colors: 55117 | B/px: 3.000
Top colors GIF (RGB - aparicions):
(99, 99, 99) - 4093
(224, 188, 41) - 3649
(48, 43, 33) - 3348
(96, 97, 73) - 2885
(27, 24, 22) - 2837
(197, 161, 30) - 2800
(103, 106, 87) - 2777
(67, 65, 52) - 2667
Bottom colors GIF (RGB - aparicions):
(26, 27, 25) - 501
(149, 120, 78) - 472
(36, 33, 31) - 387
(109, 135, 118) - 304
(186, 142, 78) - 262
(51, 45, 39) - 261
(170, 136, 82) - 238
(102, 102, 99) - 233
Top colors PNG (RGB - aparicions):
(99, 99, 99) - 3467
(0, 0, 0) - 768
(28, 27, 24) - 719
(30, 28, 26) - 694
(26, 24, 22) - 685
(24, 23, 20) - 655
(27, 25, 23) - 651

(20, 18, 19) - 618
Bottom colors PNG (RGB - aparicions):
(255, 255, 186) - 1
(255, 255, 192) - 1
(255, 255, 213) - 1
(255, 255, 218) - 1
(255, 255, 223) - 1
(255, 255, 229) - 1
(255, 255, 232) - 1
(255, 255, 245) - 1

Detall de image2.png

GIF: image2.gif | Colors: 256 | B/px: 1.000
PNG: image2.png | Colors: 63558 | B/px: 3.000
Top colors GIF (RGB - aparicions):
(34, 24, 23) - 3625
(127, 107, 88) - 3485
(99, 99, 99) - 3255
(120, 99, 81) - 3171
(100, 84, 68) - 3150
(39, 31, 28) - 3145
(92, 71, 55) - 2992
(73, 79, 47) - 2929
Bottom colors GIF (RGB - aparicions):
(48, 49, 39) - 218
(40, 25, 27) - 177
(126, 97, 59) - 172
(136, 111, 65) - 133
(68, 79, 59) - 132
(75, 87, 66) - 84
(57, 91, 70) - 33
(56, 79, 62) - 25
Top colors PNG (RGB - aparicions):
(99, 99, 99) - 2968
(34, 24, 22) - 874
(35, 25, 24) - 851
(0, 0, 0) - 768
(37, 27, 25) - 733
(33, 23, 21) - 705
(44, 37, 34) - 704
(46, 39, 36) - 701
Bottom colors PNG (RGB - aparicions):
(255, 255, 114) - 1
(255, 255, 121) - 1
(255, 255, 122) - 1
(255, 255, 131) - 1
(255, 255, 133) - 1
(255, 255, 154) - 1
(255, 255, 165) - 1
(255, 255, 178) - 1

Detall de image3.png

GIF: image3.gif | Colors: 256 | B/px: 1.000
PNG: image3.png | Colors: 45558 | B/px: 3.000

Top colors GIF (RGB - aparicions):
(255, 255, 255) - 6993
(99, 99, 99) - 4088
(129, 143, 120) - 3430
(53, 52, 52) - 2931
(120, 137, 150) - 2754
(138, 150, 127) - 2676
(85, 90, 104) - 2617
(225, 232, 211) - 2563
Bottom colors GIF (RGB - aparicions):
(246, 228, 184) - 551
(131, 145, 132) - 492
(251, 255, 247) - 460
(252, 255, 255) - 373
(52, 53, 58) - 371
(252, 252, 243) - 276
(131, 147, 134) - 93
(251, 250, 255) - 74
Top colors PNG (RGB - aparicions):
(255, 255, 255) - 6792
(99, 99, 99) - 3243
(52, 52, 52) - 887
(51, 51, 51) - 886
(0, 0, 0) - 768
(54, 54, 54) - 738
(50, 50, 50) - 563
(55, 55, 55) - 511
Bottom colors PNG (RGB - aparicions):
(255, 254, 225) - 1
(255, 254, 227) - 1
(255, 254, 229) - 1
(255, 254, 230) - 1
(255, 254, 247) - 1
(255, 255, 199) - 1
(255, 255, 200) - 1
(255, 255, 204) - 1

Detall de image4.png

GIF: image4.gif | Colors: 256 | B/px: 1.000
PNG: image4.png | Colors: 25567 | B/px: 3.000
Top colors GIF (RGB - aparicions):
(99, 99, 99) - 4262
(181, 183, 143) - 3145
(75, 77, 54) - 2870
(169, 168, 134) - 2760
(176, 175, 137) - 2744
(173, 176, 141) - 2612
(158, 154, 120) - 2519
(183, 182, 143) - 2507
Bottom colors GIF (RGB - aparicions):
(104, 104, 101) - 299
(251, 255, 235) - 263
(89, 70, 50) - 252
(99, 108, 108) - 178

(173, 170, 140) - 159
(172, 167, 140) - 123
(179, 184, 132) - 48
(177, 169, 117) - 16
Top colors PNG (RGB - aparicions):
(99, 99, 99) - 3225
(177, 176, 139) - 2267
(179, 177, 140) - 2110
(176, 175, 137) - 2086
(169, 168, 133) - 1955
(171, 169, 134) - 1869
(168, 166, 131) - 1831
(175, 173, 136) - 1683
Bottom colors PNG (RGB - aparicions):
(255, 233, 213) - 1
(255, 242, 212) - 1
(255, 245, 212) - 1
(255, 248, 221) - 1
(255, 252, 223) - 1
(255, 255, 211) - 1
(255, 255, 219) - 1
(255, 255, 220) - 1

Detall de image5.png

GIF: image5.gif | Colors: 256 | B/px: 1.000
PNG: image5.png | Colors: 24470 | B/px: 3.000
Top colors GIF (RGB - aparicions):
(255, 255, 255) - 49555
(255, 255, 224) - 4340
(255, 255, 239) - 3677
(255, 255, 243) - 3629
(255, 255, 223) - 3493
(255, 255, 225) - 3461
(255, 255, 240) - 3300
(255, 255, 249) - 3257
Bottom colors GIF (RGB - aparicions):
(204, 181, 139) - 188
(239, 221, 168) - 167
(119, 102, 57) - 164
(104, 103, 73) - 161
(135, 110, 47) - 123
(254, 234, 190) - 122
(129, 108, 55) - 42
(249, 254, 240) - 3
Top colors PNG (RGB - aparicions):
(255, 255, 255) - 49555
(255, 255, 224) - 4237
(255, 255, 239) - 3677
(255, 255, 243) - 3629
(255, 255, 223) - 3493
(255, 255, 225) - 3403
(255, 255, 240) - 3300
(255, 255, 249) - 3257
Bottom colors PNG (RGB - aparicions):

(255, 255, 176) - 1
(255, 255, 179) - 1
(255, 255, 185) - 1
(255, 255, 187) - 1
(255, 255, 192) - 1
(255, 255, 194) - 1
(255, 255, 198) - 1
(255, 255, 200) - 1

Detall de image6.png

GIF: image6.gif | Colors: 256 | B/px: 1.000
PNG: image6.png | Colors: 72079 | B/px: 3.000
Top colors GIF (RGB - aparicions):
(225, 214, 227) - 3588
(194, 184, 172) - 3480
(105, 97, 73) - 3237
(219, 174, 1) - 3178
(98, 90, 66) - 3134
(94, 153, 163) - 3073
(101, 93, 70) - 2981
(164, 163, 170) - 2954
Bottom colors GIF (RGB - aparicions):
(64, 55, 46) - 314
(82, 122, 85) - 280
(68, 57, 41) - 244
(245, 203, 161) - 200
(226, 185, 102) - 199
(85, 104, 75) - 172
(212, 157, 91) - 162
(248, 226, 168) - 30
Top colors PNG (RGB - aparicions):
(65, 59, 46) - 1240
(255, 255, 255) - 1106
(66, 61, 47) - 972
(63, 58, 45) - 957
(65, 59, 48) - 934
(0, 0, 0) - 768
(66, 60, 50) - 736
(94, 92, 77) - 703
Bottom colors PNG (RGB - aparicions):
(255, 253, 242) - 1
(255, 254, 221) - 1
(255, 254, 223) - 1
(255, 254, 233) - 1
(255, 254, 237) - 1
(255, 254, 252) - 1
(255, 255, 227) - 1
(255, 255, 228) - 1

Detall de image7.png

GIF: image7.gif | Colors: 256 | B/px: 1.000
PNG: image7.png | Colors: 258 | B/px: 3.000
Top colors GIF (RGB - aparicions):
(255, 255, 255) - 312141

(0, 0, 0) - 19685
(254, 254, 254) - 2575
(1, 1, 1) - 1059
(2, 2, 2) - 833
(253, 253, 253) - 820
(3, 3, 3) - 709
(4, 4, 4) - 671
Bottom colors GIF (RGB - aparicions):
(161, 161, 161) - 144
(166, 166, 166) - 144
(132, 132, 132) - 143
(174, 174, 174) - 142
(123, 123, 123) - 141
(138, 138, 138) - 140
(147, 147, 147) - 139
(148, 148, 148) - 138
Top colors PNG (RGB - aparicions):
(255, 255, 255) - 312141
(0, 0, 0) - 19685
(254, 254, 254) - 2575
(1, 1, 1) - 1059
(2, 2, 2) - 833
(253, 253, 253) - 820
(3, 3, 3) - 709
(4, 4, 4) - 671
Bottom colors PNG (RGB - aparicions):
(132, 132, 132) - 143
(174, 174, 174) - 142
(123, 123, 123) - 141
(147, 147, 147) - 139
(138, 138, 138) - 138
(148, 148, 148) - 138
(137, 138, 138) - 2
(139, 140, 140) - 1

Detall de image8.png

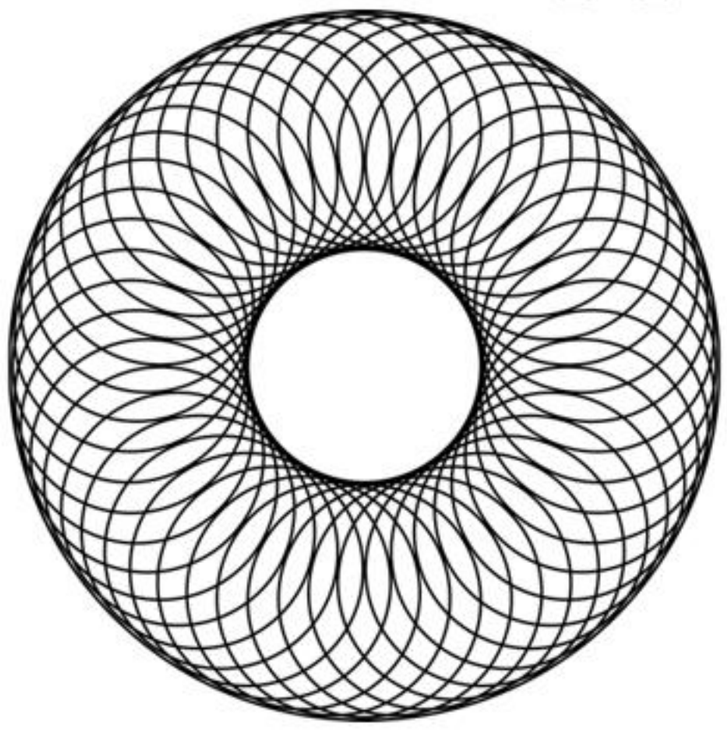
GIF: image8.gif | Colors: 244 | B/px: 1.000
PNG: image8.png | Colors: 2377 | B/px: 3.000
Top colors GIF (RGB - aparicions):
(254, 146, 1) - 49566
(102, 190, 236) - 49542
(255, 229, 0) - 49515
(160, 92, 255) - 49023
(0, 113, 223) - 48402
(223, 16, 0) - 47718
(56, 167, 0) - 47282
(56, 167, 1) - 1116
Bottom colors GIF (RGB - aparicions):
(255, 228, 0) - 15
(102, 192, 237) - 14
(99, 192, 234) - 13
(102, 190, 238) - 12
(221, 11, 1) - 10
(104, 190, 237) - 5

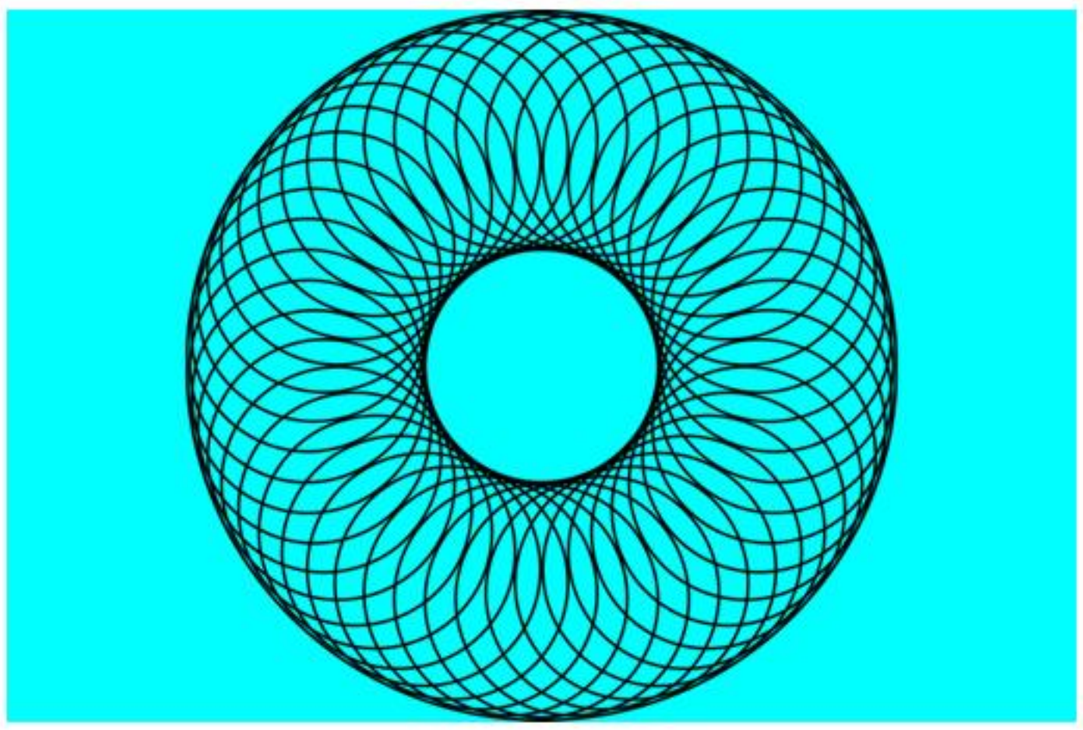
(0, 110, 222) - 2
(253, 145, 0) - 2
Top colors PNG (RGB - aparicions):
(254, 146, 1) - 49566
(255, 229, 0) - 49515
(102, 190, 236) - 49505
(160, 92, 255) - 49023
(0, 113, 223) - 48402
(223, 16, 0) - 47718
(56, 167, 0) - 47277
(56, 167, 1) - 1116
Bottom colors PNG (RGB - aparicions):
(253, 139, 3) - 1
(253, 149, 3) - 1
(253, 154, 1) - 1
(254, 181, 2) - 1
(254, 186, 2) - 1
(254, 212, 1) - 1
(254, 227, 1) - 1
(255, 212, 1) - 1

Apartat g - Canvi de paleta (R=0)

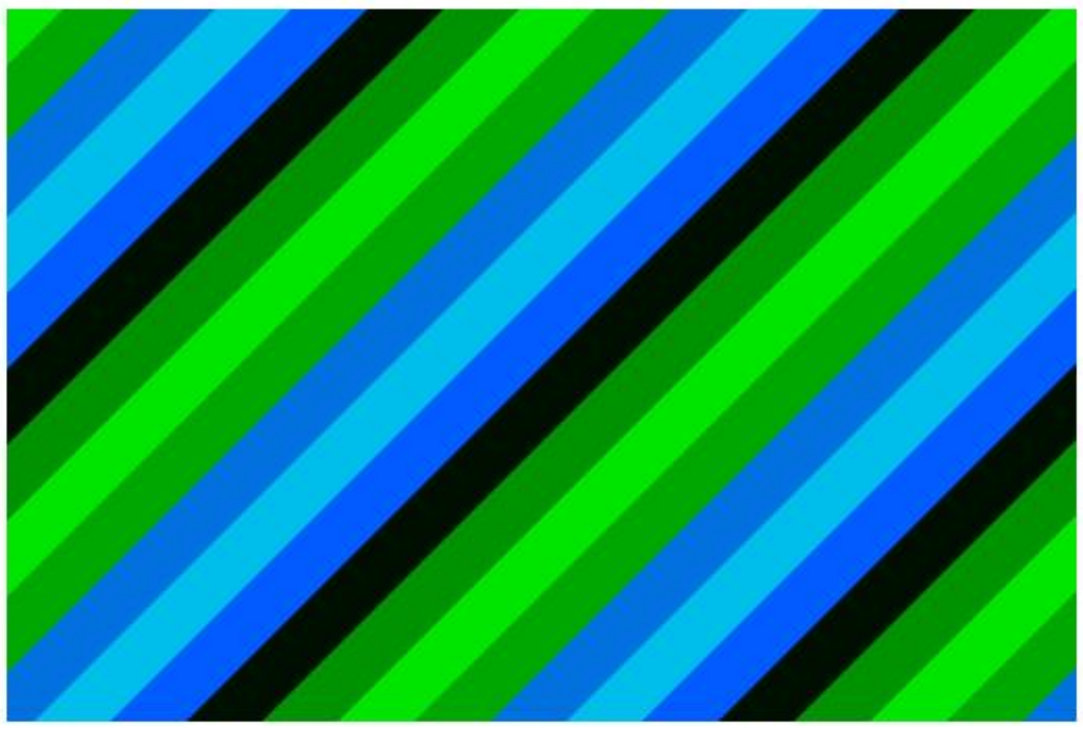
A continuacio es mostren les comparacions visuals entre el GIF original i el GIF modificat amb component vermella anul-lada (R=0) per a les imatges 3, 7 i 8.

Esquerra: image3.gif | Dreta: image3_no_red.gif





Esquerra: image7.gif | Dreta: image7_no_red.gif



Esquerra: image8.gif | Dreta: image8_no_red.gif